

Vector Space, Sub Space, Basis & Span :

Introduction to Vector space, Subspace, Basis & Span

* Vector space

Linear combination $\begin{cases} \swarrow \text{Scalar multiplication} \\ \searrow \text{Vector Addition} \end{cases}$

→ Vector space is set (collection) of vectors such that their Linear Combination will give a vector from the same set.

→ $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4 \dots \mathbb{R}^n$ are examples of vector space

→ $\mathbb{R}^2$ means set of all possible column vectors v with 2 components. $\rightarrow \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$

Linear dependence applies to a set of vectors *within* a pre-existing vector space, meaning **the vector space must exist first for linear dependence to be defined.**

→ $\mathbb{R}^3$ means set of all possible column vectors v with 3 components.

$$\rightarrow \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

→ $\mathbb{R}^n$ means set of all possible column vectors v with n components.

$$\rightarrow \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

→ Vector space should contain zero vector.

$$0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \text{ (additive identity)}$$

→ Closed under linear combination.

For any vectors u, v in V and scalars a, b ,

$$a \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + b \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \in V \quad (c \text{ is also in the same vector space})$$

$$\text{i.e., } a, b \in \mathbb{R}^n \Rightarrow a + b = c \in \mathbb{R}^n$$

→ All vectors belong to one of the space $\mathbb{R}^n$ where $n = \text{no of components in } v$

* Subspace of vector space

→ A subspace of a vector space is a non-empty subset that satisfies requirements of vector space

↳ Linear combinations stay in the subspace

$$\text{Subspace} = \text{Subset} + \text{Vector space}$$

→ Or simply, A subspace is a subset that is closed under Addition and scalar multiplication.

→ Zero vector should always belong to the subspace.

Example: For $\mathbb{R}^3$ being a vector space

① A set containing only zero vector ← Subspace

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② Line through origin (1 LI vector) ← Subspace

③ Plane through origin (2 LI vector) ← Subspace

④ Plane not through origin ← not subspace

dimension of vector
Space = no of LI
vectors

Example: For $\mathbb{R}^2$ being a vector space

Keep only the vectors $\begin{bmatrix} x \\ y \end{bmatrix}$ where $(x > 0, y > 0)$ or $(x < 0, y < 0)$ or $(x = 0, y = 0)$.
Is this a subspace?

→ Subspace = Subset + vector space

Let's use proof by counterexample

$$u = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } v = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$$u + v = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \notin \mathbb{R}^2 \text{ so not a subspace of } \mathbb{R}^2$$

There is a mistake in the above sentence that it will belong to $\mathbb{R}^2$; but not a subspace.

* vector space & Linear Independence

→ Max no of Linearly Independent vectors in $\mathbb{R}^2 \rightarrow 2$

Any 2 Linearly Independent vectors in $\mathbb{R}^2$, will fill the space in $\mathbb{R}^2$

Example: which of the below vectors can fill the $\mathbb{R}^2$ vector space?

✓ ① $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

✓ ② $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$

✓ ③ $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

→ Max no of Linearly Independent vectors in $\mathbb{R}^n \rightarrow n$

Any 2 Linearly Independent vectors in $\mathbb{R}^2$, will fill the space in $\mathbb{R}^2$

* Span of a vector set

→ Span of a vector set is the set of all possible linear combination of vectors

↓
weighted sum

→ Mathematically:

If $v_1, v_2, v_3 \dots v_k \in \mathbb{R}^n$

then $\text{span}(\{v_1, v_2, v_3 \dots v_k\}) = \{a_1 v_1 + a_2 v_2 + \dots + a_k v_k \mid a_i \in \mathbb{R}\}$

Example : Find span of

① $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\} \leftarrow \text{a line}$

② $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right\} \leftarrow \text{a line}$

③ $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\} \leftarrow \text{a plane}$

④ $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \end{bmatrix} \right\} \leftarrow \text{a plane}$

⑤ $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \leftarrow \text{a plane}$
 $\underbrace{\hspace{1.5cm}}_{2 \perp \text{ vectors?}}$

⑥ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \leftarrow \text{a 3d space}$

⑦ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \right\} \leftarrow \text{a plane}$

Example: Mathematically express the span of

$$\textcircled{1} \text{ span} \left(\underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{v_2} \right)$$

$$\rightarrow \text{Linear Combination} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$$

$$\text{span}(\{v_1, v_2\}) = \left\{ \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

$$\textcircled{2} \text{ span} \left(\underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{v_1}, \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{v_2} \right)$$

$$\rightarrow \text{Linear Combination} = a \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ b \\ a \end{bmatrix}$$

$$\text{span}(\{v_1, v_2\}) = \left\{ \begin{bmatrix} 0 \\ b \\ a \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

* Basis of a vector space

$\rightarrow$ A basis of a vector space is a set of vectors that

① fills (spans) the space

② is Linearly Independent

Example: which of the following is a basis of $\mathbb{R}^2$

Example: which of the following is a basis of $\mathbb{R}^3$

① $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$

→ 3 LI vectors

✓ ② $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

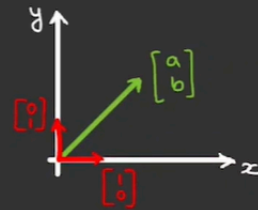
③ $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} a \\ b \\ c \end{bmatrix} \right\}$

* Standard basis

→ A special basis where each vector has one component is 1 and all other is 0, is standard basis.

→ standard basis of $\mathbb{R}^2 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} a \\ b \end{bmatrix} = a \hat{i} + b \hat{j} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



→ standard basis of $\mathbb{R}^3 = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \hat{i} + b \hat{j} + c \hat{k} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

